



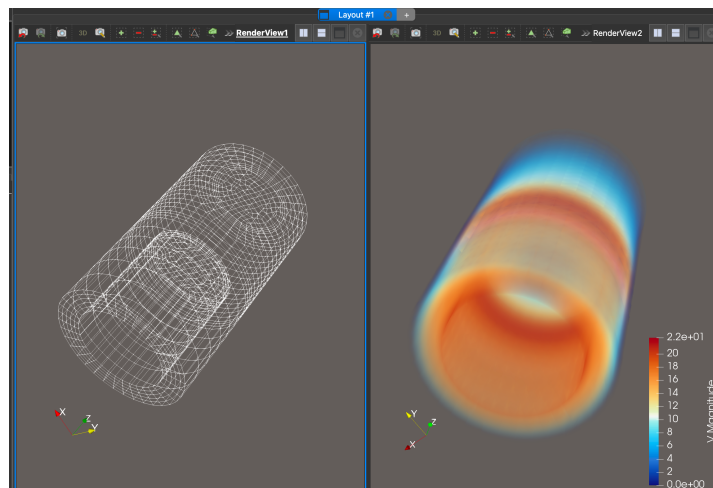
Tutorial 04 – Paraview vector fields visualization

Open the `disk_out_ref.ex2` data set.

- Select **Apply**
- Set the colouring to **Solid Color** and **Wireframe**

A brief explanation of the data set: the cylinder mesh you can see is the data domain. Air enters this domain at the top Z base with velocity direction -Z. The cylinder hole in the bottom Z half of the data domain corresponds to a fast-rotating hot disk; it cannot be rendered because its geometry has not been meshed and included in the data set. The data set includes a number of properties of air within the domain, such as pressure, temperature and velocity (vector field).

The image below presents two views of the data set: the wireframe mesh and a volume rendering of the velocity magnitude.



Direct Vector Visualization

Let's directly visualize the velocity vector field by depicting it with arrows.

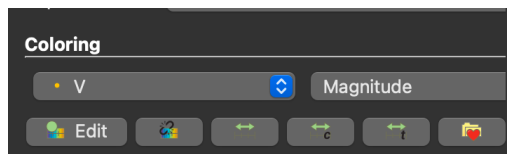
- Select **Filters** → **Common** → **Glyph** (or press )

On the Properties panel:

- Make sure **Glyph Type** is set to **Arrow**
- Set **Orientation Array** to **V**
- Set **Scale Array** to **No Scale Array**
- Set **Scale Factor** to **0.8**
- Press **Apply**
- Make sure the **disk_out_ref.ex2** pipeline stage is not selected for display, such that the wireframe is not presented

We can do a little better:

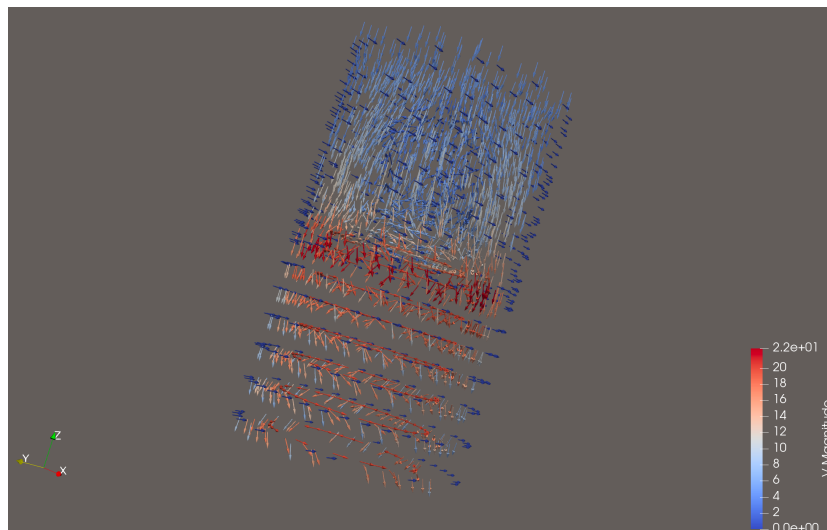
- Set **Coloring** to **V**:



- Press **Edit** and then change the colormap to “**Cool to Warm**”

Explore the visualization:

1. If you look along the positive Z-axis, in which direction is the disk rotating: counterclockwise or clockwise?
2. If you look along the positive X-axis can you see any air flow perturbation above the rotating disk?
3. Where does the air flow with larger linear velocity? Near the center or near the border of the rotating disk top? Why?

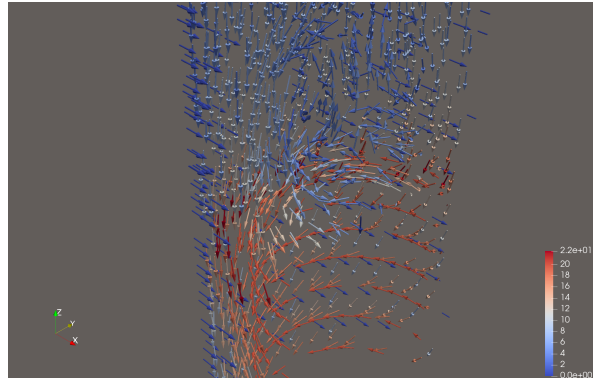


Let's clip the data, cutting the data set in half along a XZ plane, in order to better visualize what is going on inside:

- With **Glyph1** selected add a **Clip** plane ()
- Select “**Y Normal**” and then press **Apply**

Make sure only the Clip1 filter is selected for display. The data set was cut along the XZ plane, allowing visualization of the vector field inside the volume.

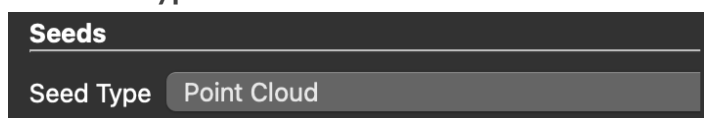
Play around with the camera position. You should be able to identify air going up on top of the spinning hot disc and the spinning of air in the top and around this same disc.



Streamlines : Indirect Vector Visualization

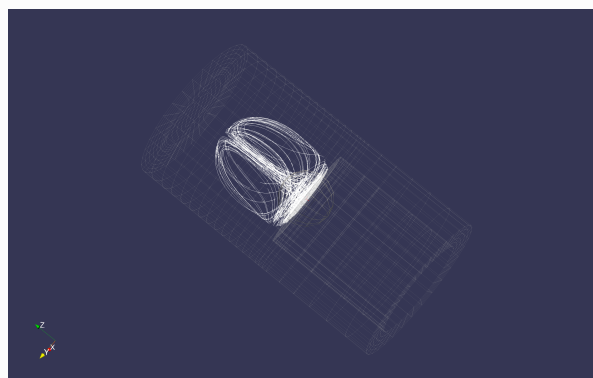
We can probably do much better to visualize the air flow within the domain.

- Reset Paraview : 
- Open the `disk_out_ref.ex2` data set.
- Press **Apply**
- Set the colouring to **Solid Color** and **Wireframe**
- Add a **StreamTracer** filter ()
- Set **Seed Type** to **Point Cloud**

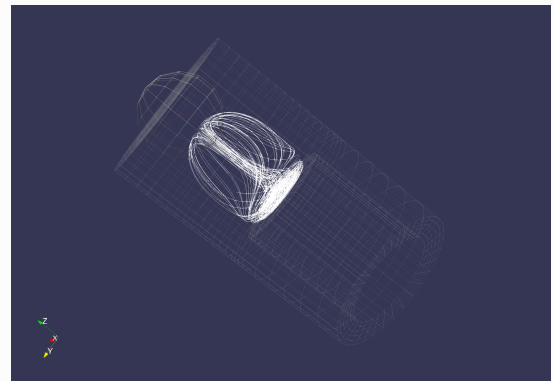
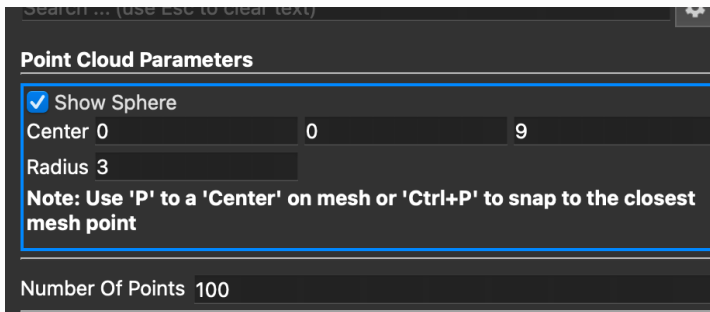


- Press **Apply**

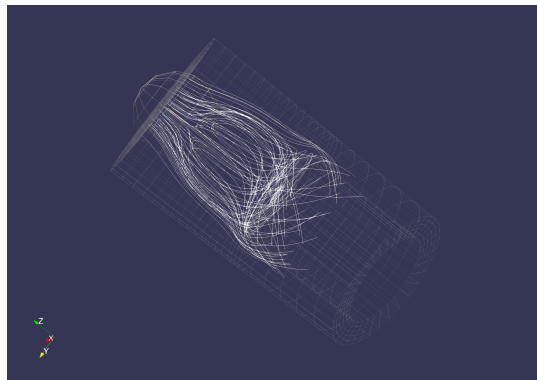
You'll see that the sphere used to select seed points is at the center of the data domain:



Let us bring it up to where air enters the domain and set its radius such that it covers a significant fraction of the top:

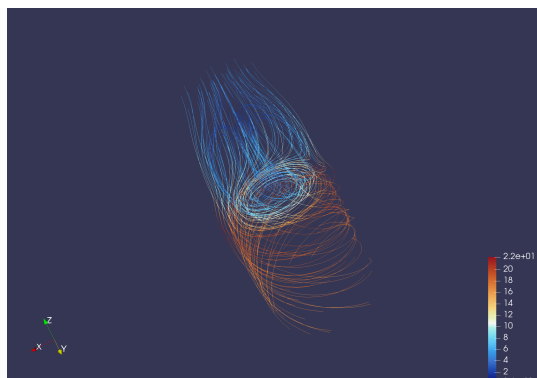


- Press **Apply**



It is looking much more informative now. Let's improve the visualization:

- Set “**Maximum StreamLine Length**” to 50
- Set “**Number of Points**” to 100
- Set “**Coloring**” to V
- Press **Apply**



Let's increase the information carried by the streamlines by shading them:

- Change the **Line Width** to 4:

Styling

Line Width 4

- Set **Render Lines as Tubes**:

Styling

☒ Render Lines As Tubes

Information on the direction of the streamlines is missing. Let's add glyphs as cones:

- Select **Filters** → **Common** → **Glyph** , or press 
- Set **Glyph Type** to **Cone**
- Set the orientation of the cones to be the velocity **V**

Orientation

Orientation Array • V

- No Scale Array.
- Scale 0.2
- Set **Glyph Mode** to:

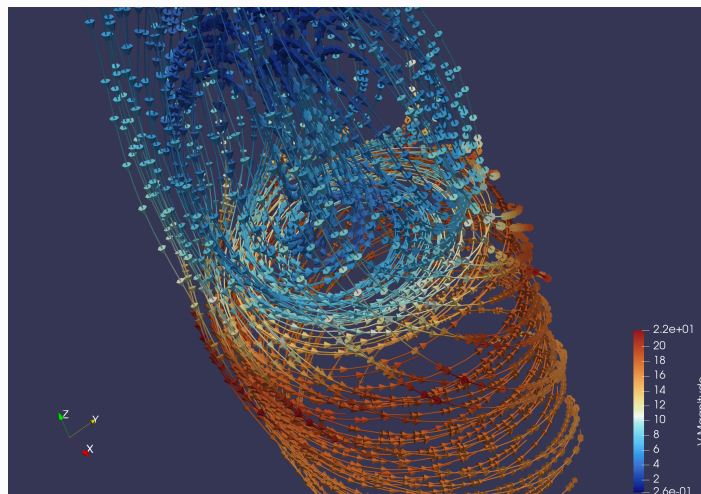
Masking

Glyph Mode Every Nth Point

Stride 2

- Set **Coloring** to **V**
- **Apply**.

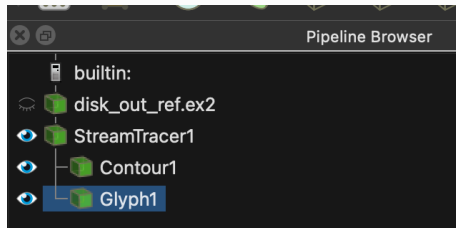
The flow of air inside the data domain should now be much clearer:



Let's filter the output of the StreamTracer such that the data refers to some value of the integration time. Make sure **StreamTracer1** is selected in the **Pipeline Browser**:

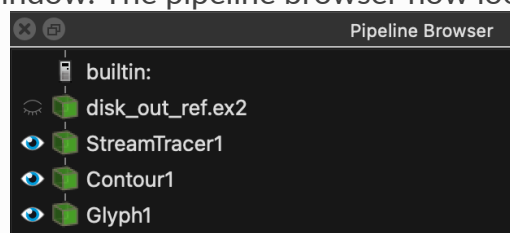
- Select **Filters** → **Common** → **Contour** , or press 
- Contour on **IntegrationTime**.
- **Apply**.

Your pipeline should look like this:

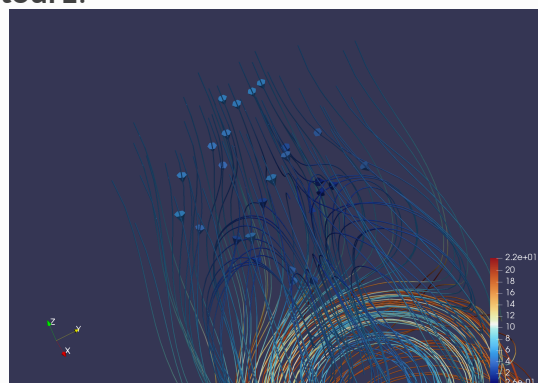


So **Glyph1** is getting its input from **StreamTracer1**. We want its input to come from **Contour1**.

Click on **Glyph1** with the mouse right button, select “**Change Input**” and then select **Contour1** on the Dialog window. The pipeline browser now looks like:



If you look at the rendered image, cones only appear over the points of the streamlines currently allowed by **Contour1**:



Does this suggest how to build an animation of the cones along the streamlines?

Generating an animation from the velocity static vector field

This data set does not contain multiple time instants, but since $\mathbf{V}$ is a vectorial field the Stream Tracer filter, traces the velocity over a time interval. In fact, if you inspect the **Contour1 Value Range** you can see the time interval values:

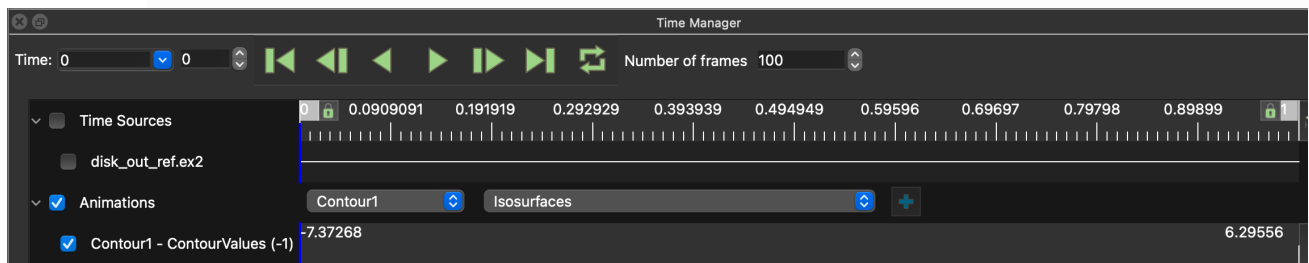
Isosurfaces

Value Range: [-7.37268, 6.29556]

So an animation of the glyphs (cones) can be made by changing over time (animation time) the integration time the Contour Filter is passing onto the Glyph filter.

Let's open the Time Manager window:

- Select **View → Time Manager**.
- Since there are no time instants in the data uncheck **Time Sources**.
- Change the pulldown box next to **Animations** to be **Contour1**.
- Click the blue +. Note that the time interval is set to be the same as the value range in the contour by time integration. Change the initial time to -0.4 and final time to 5.5, just because the animation will look more interesting
- Set **No. Frames** to 200.



- Now, click the play button.

Save this as a movie.

- **File → Save Animation.**
- Add a file name.
- Save as a **.avi**.
- **OK.**
- **OK.**